

Sri Lanka Institute of Information Technology



Data Warehousing and Business Intelligence (IT3021) Assignment 2

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Step 1. Introduction of the data set

Background

The Central table, fact_sales, stores measurable transactional data such as the quantity sold and sales amount. It also contains foreign keys that reference the dimension tables, allowing integration of descriptive attributes with transactional data.

The dimensional tables include Dim_customer, Dim_product, Dim_dealsize and Dim_date. Each dimensional table contains descriptive information related to customers, products, time and deal size respectively. Surrogate keys are used as primary keys in each dimension table to uniquely identify records and improve joint performance.

The fact table is connected to each dimension table through foreign key relationships forming a star schema.



Figure 1.1 Star schema

1.1 Fact and dimension tables

Column Name	Data Type	Description
sales_id (PK)	INT	Unique transaction identifier
customer_id (FK)	INT	Links to Dim Customer
product_id (FK)	INT	Links to Dim Product
date_id (FK)	INT	Links to Dim Date
dealsize_id (FK)	INT	Links to Dim Dealsize
quantity_sold	INT	Number of items sold in the transaction
sales_amount	DECIMAL(10,2)	Total sales amount
acm_txn_create_time	DATETIME	Transaction creation time
acm_txn_complete_time	DATETIME	Transaction completion time
txn_process_time_hours	DECIMAL(5,2)	Time difference between creation

		and completion in hours
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Table Name	Column Name	Data Type	Description
Dim_Date	date_id (PK)	INT	Unique identifier for each date
Dim_Date	full_date	DATE	Actual date value
Dim_Date	month	INT	Month number (1–12)
Dim_Date	quarter	INT	Quarter of the year (1–4)
Dim_Date	year	INT	Year value
Dim_Customer	customer_id (PK)	INT	Unique identifier for each customer
Dim_Customer	customer_name	VARCHAR(100)	Name of the customer
Dim_Customer	city	VARCHAR(50)	City of the customer
Dim_Customer	country	VARCHAR(50)	Country of the customer
Dim_Customer	customer_segment	VARCHAR(50)	Customer category (Small, Medium, Large)
Dim_Customer	age_group	VARCHAR(50)	Age category (Young, Adult, Senior)
Dim_Product	product_id (PK)	INT	Unique identifier for each product
Dim_Product	product_code	VARCHAR(50)	Original product code
Dim_Product	product_line	VARCHAR(50)	Product line category
Dim_Product	category	VARCHAR(50)	High-level product category
Dim_Product	subcategory	VARCHAR(50)	Detailed product classification
Dim_Dealsize	dealsize_id (PK)	INT	Unique identifier for deal size
Dim_Dealsize	deal_size	VARCHAR(50)	Deal size category (Small, Medium, Large)

1.2 ER diagram

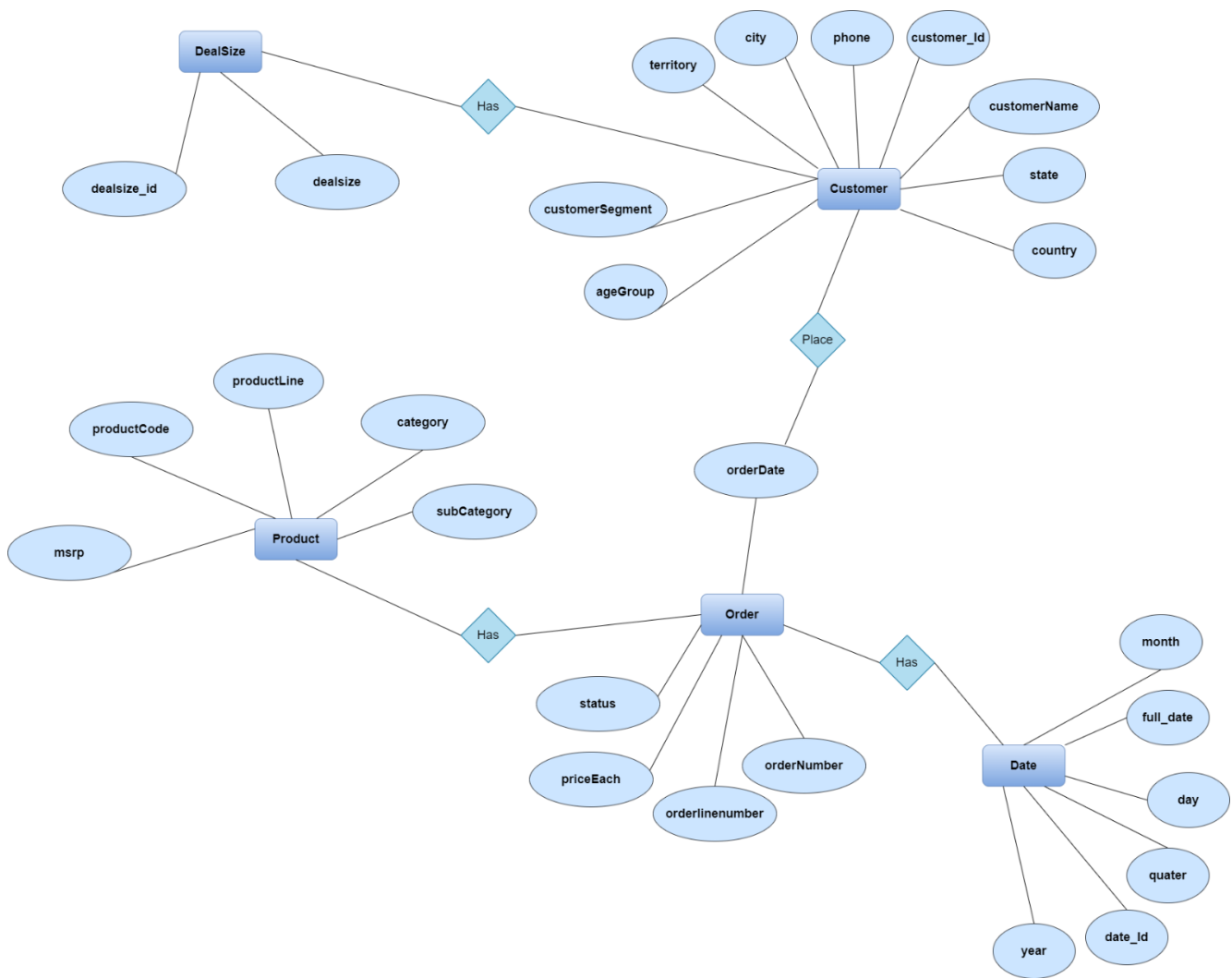
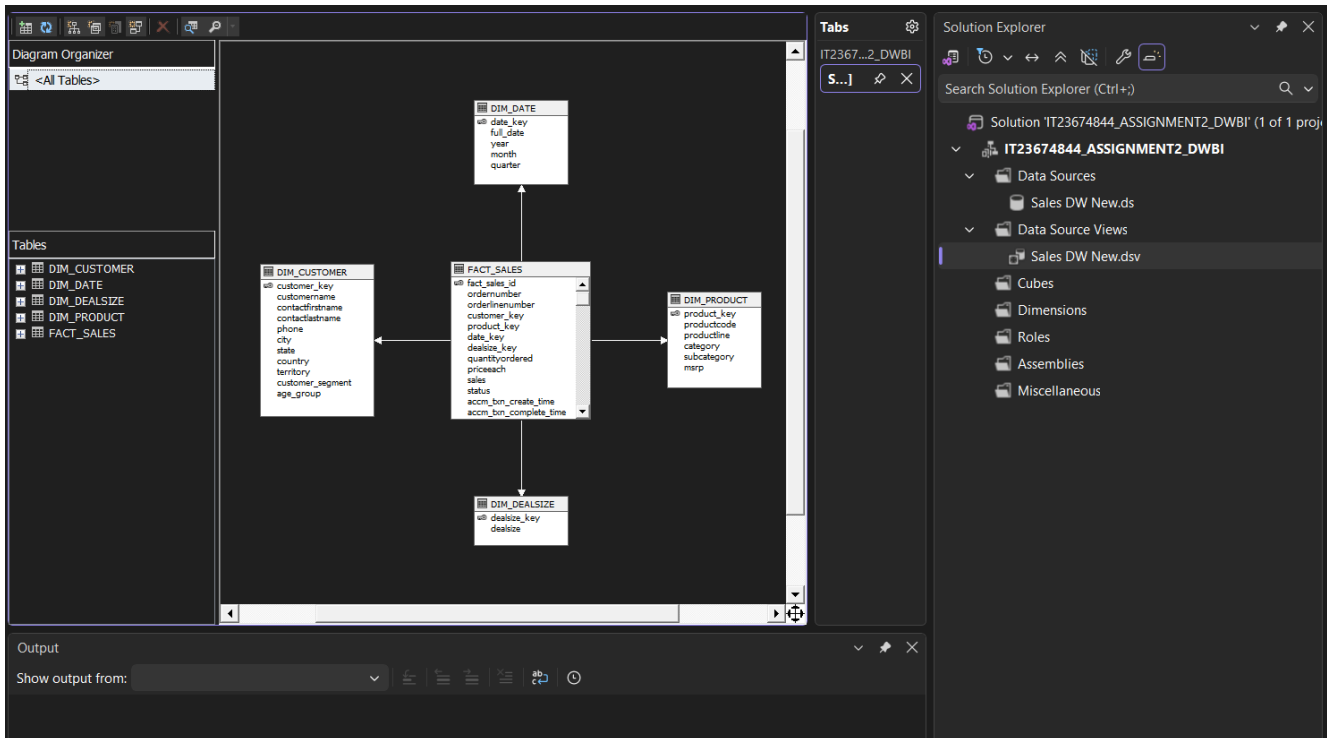


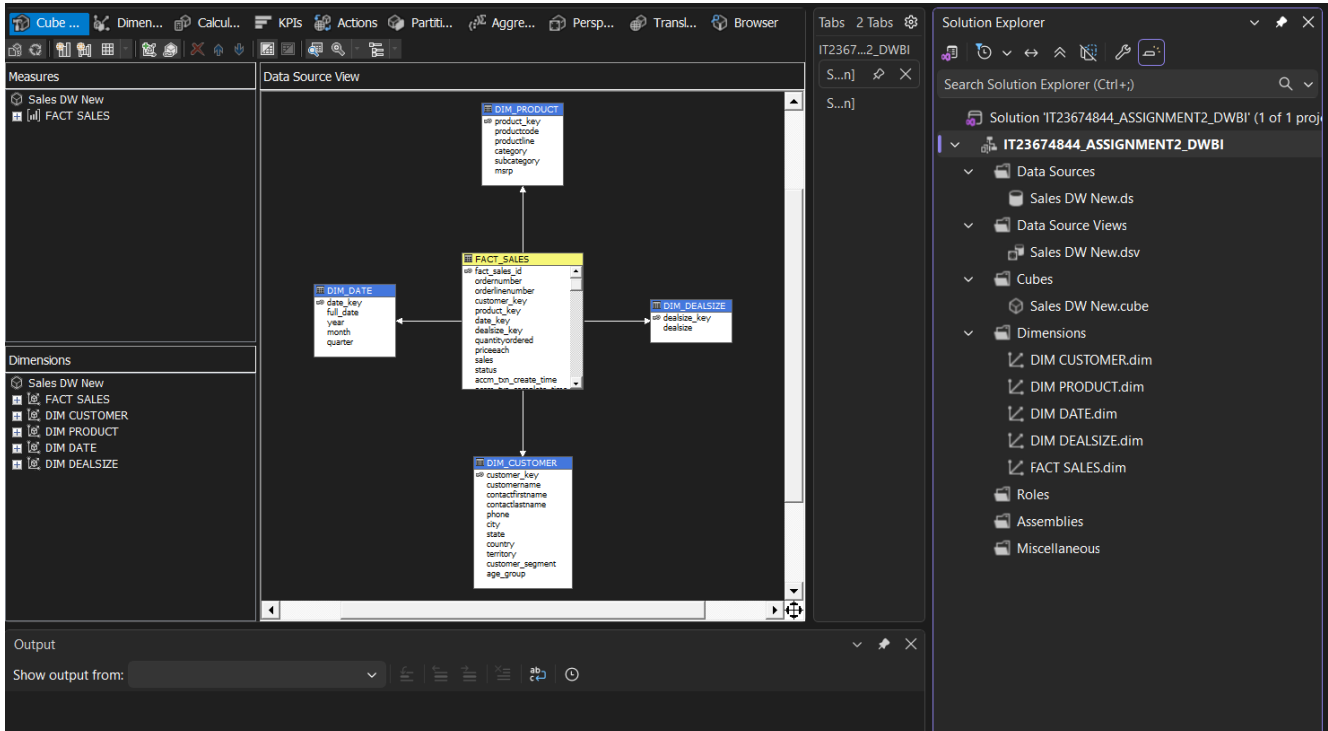
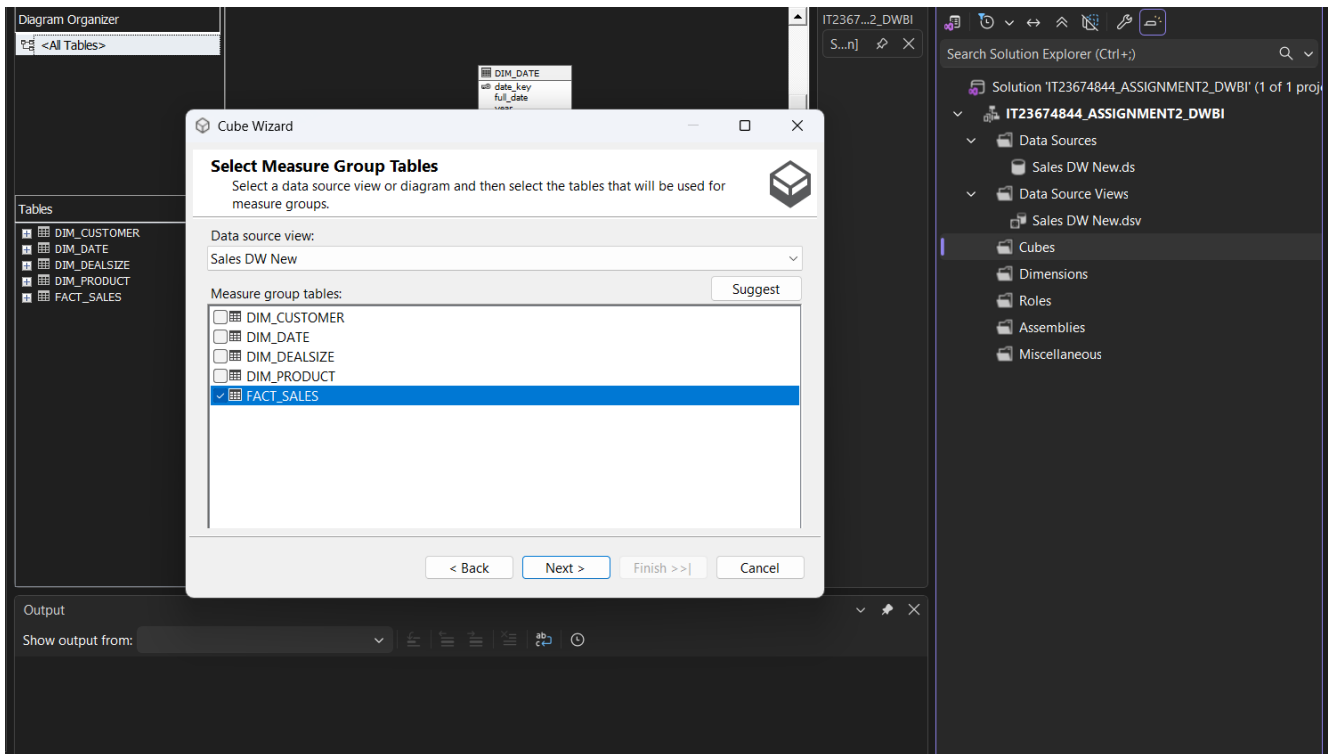
Figure 1..2 ER diagram

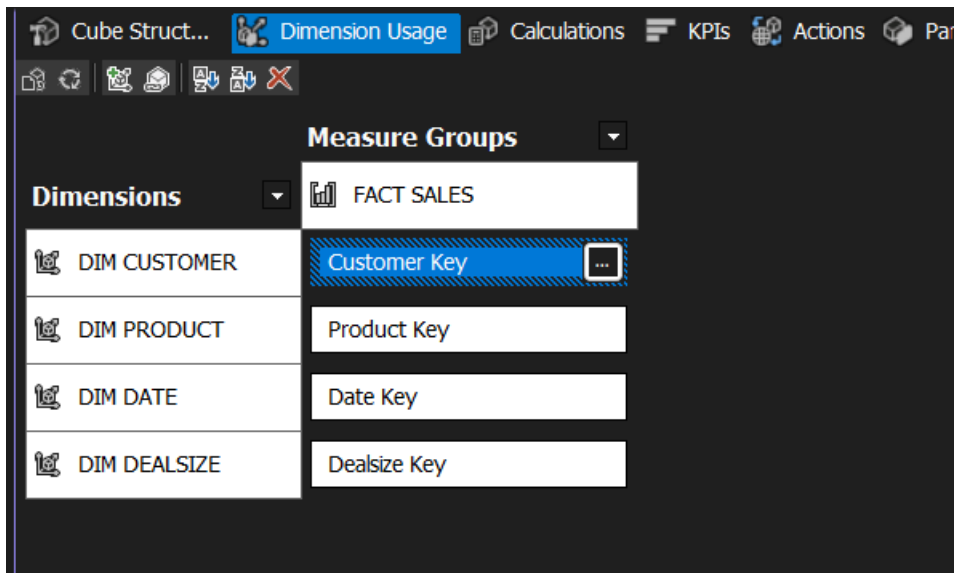
Step 2 . Implementation of the SSAS Cube

2.1 Steps followed

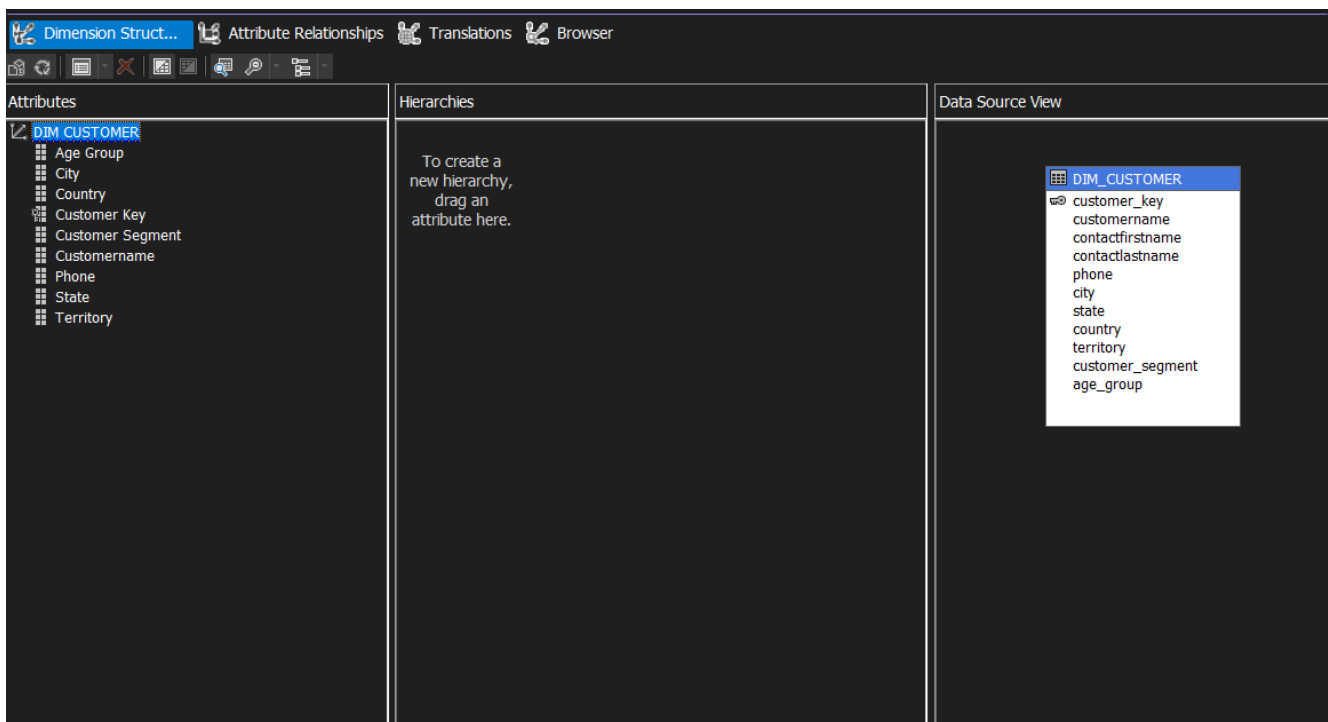
An SSAS multidimensional cube was developed using SQL server data tools (SSDT) by connecting to the data warehouse as the primary data source. A data source and data source view were created to include the necessary fact and dimension tables. Dimensions such as product, customer, date and deal size within defined and relevant attributes were selected for analysis. A cube was created using the fact sales table and key measures such as sales quantity ordered were added. Relationships between the cube and dimensions were established to enable multi dimensional analysis.







2.1.1 Customer dimension



2.1.2 Product dimension

The screenshot shows the 'Dimension Structure' tool interface. The top menu bar includes 'Dimension Struct...', 'Attribute Relationships', 'Translations', and 'Browser'. Below the menu is a toolbar with various icons. The main workspace is divided into three panels:

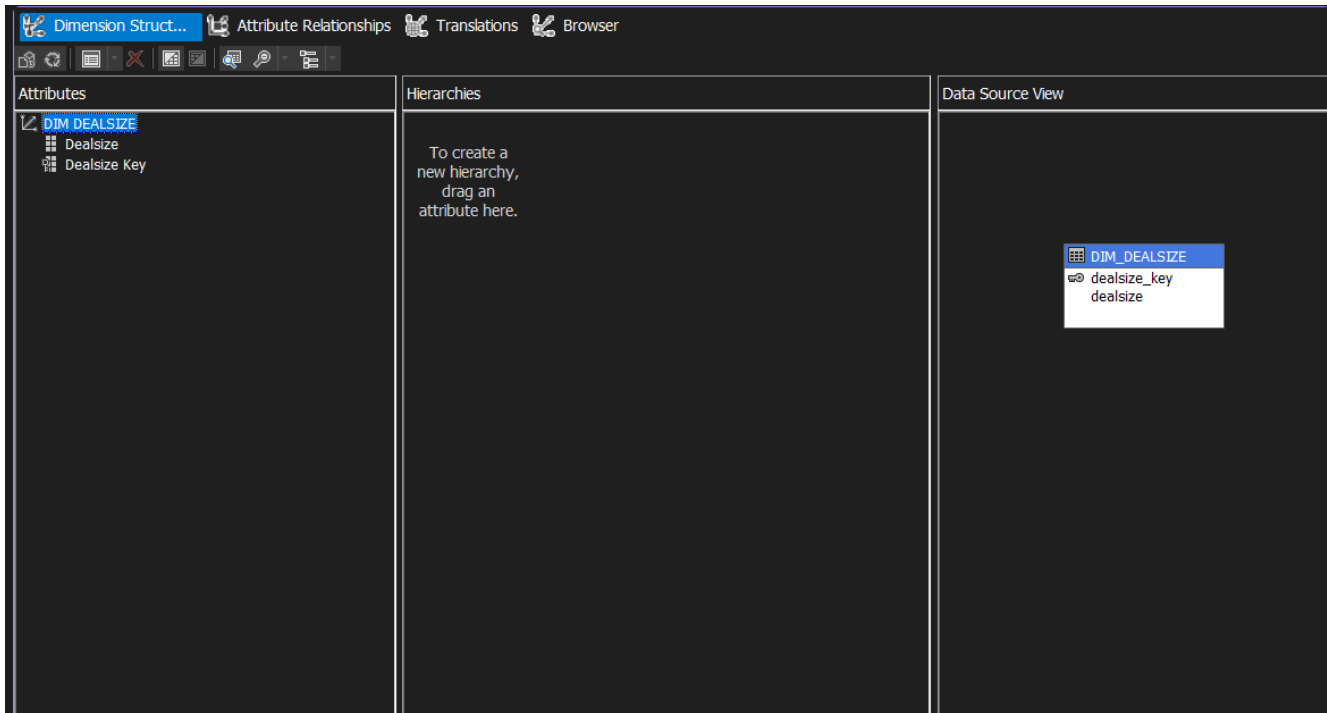
- Attributes:** Lists attributes for 'DIM_PRODUCT': Category, Msrp, Product Key, Productcode, Productline, and Subcategory.
- Hierarchies:** Shows a hierarchy for 'product_Hierarchy' with levels: Category, Subcategory, Productline, and <new level>. A tooltip indicates: 'To create a new hierarchy, drag an attribute here.'
- Data Source View:** Displays the 'DIM_PRODUCT' data source with attributes: product_key, productcode, productline, category, subcategory, and msrp.

2.1.3 Date dimension

The screenshot shows the 'Dimension Structure' tool interface for the Date dimension. The top menu bar includes 'Dimension Struct...', 'Attribute Relationships', 'Translations', and 'Browser'. Below the menu is a toolbar with various icons. The main workspace is divided into three panels:

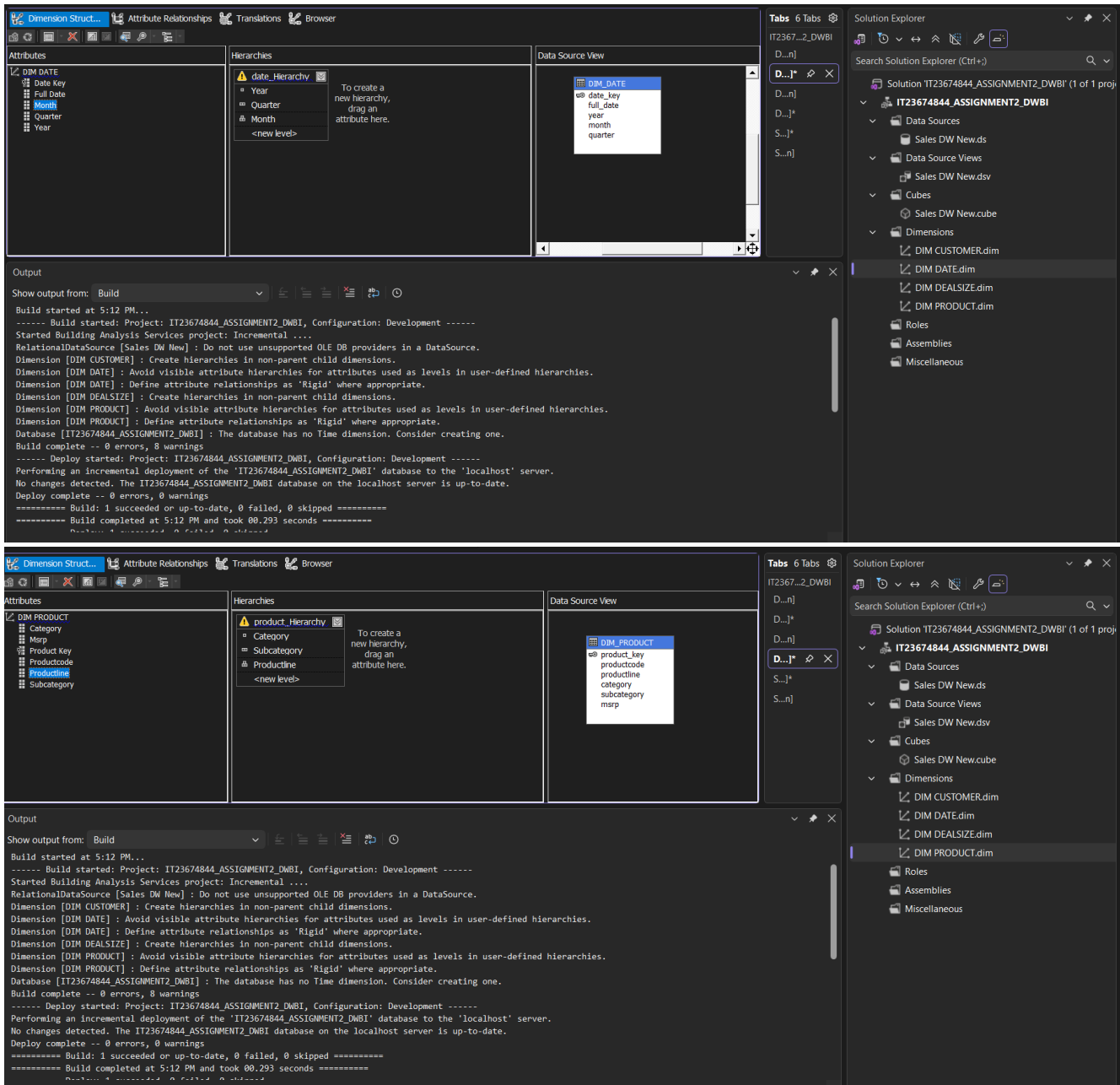
- Attributes:** Lists attributes for 'DIM_DATE': Date Key, Full Date, Month, Quarter, and Year.
- Hierarchies:** Shows a hierarchy for 'date_Hierarchy' with levels: Year, Quarter, and Month, followed by <new level>. A tooltip indicates: 'To create a new hierarchy, drag an attribute here.'
- Data Source View:** Displays the 'DIM_DATE' data source with attributes: date_key, full_date, year, month, and quarter.

2.1.4 Deal Size dimension



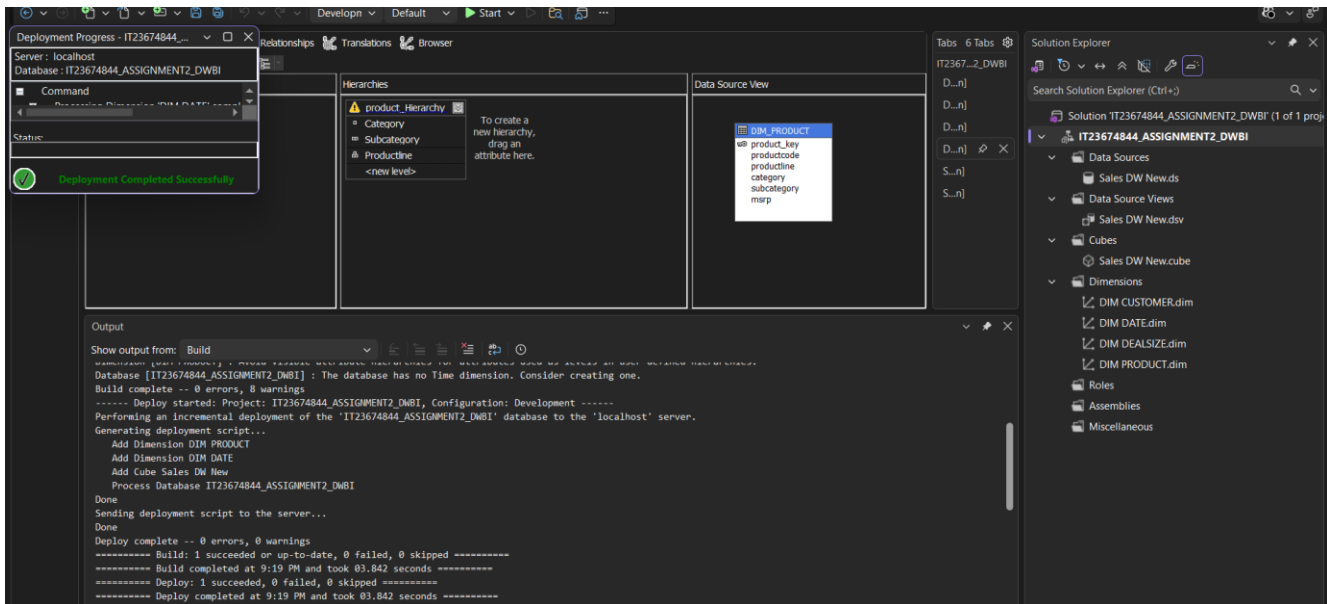
2.2 Hierarchy Implementation

Additionally, a hierarchy was implemented within the date dimension and product dimension organizing data into year, quarter, month levels the end category, sub category, product line levels. Allowing efficient drill down operations.



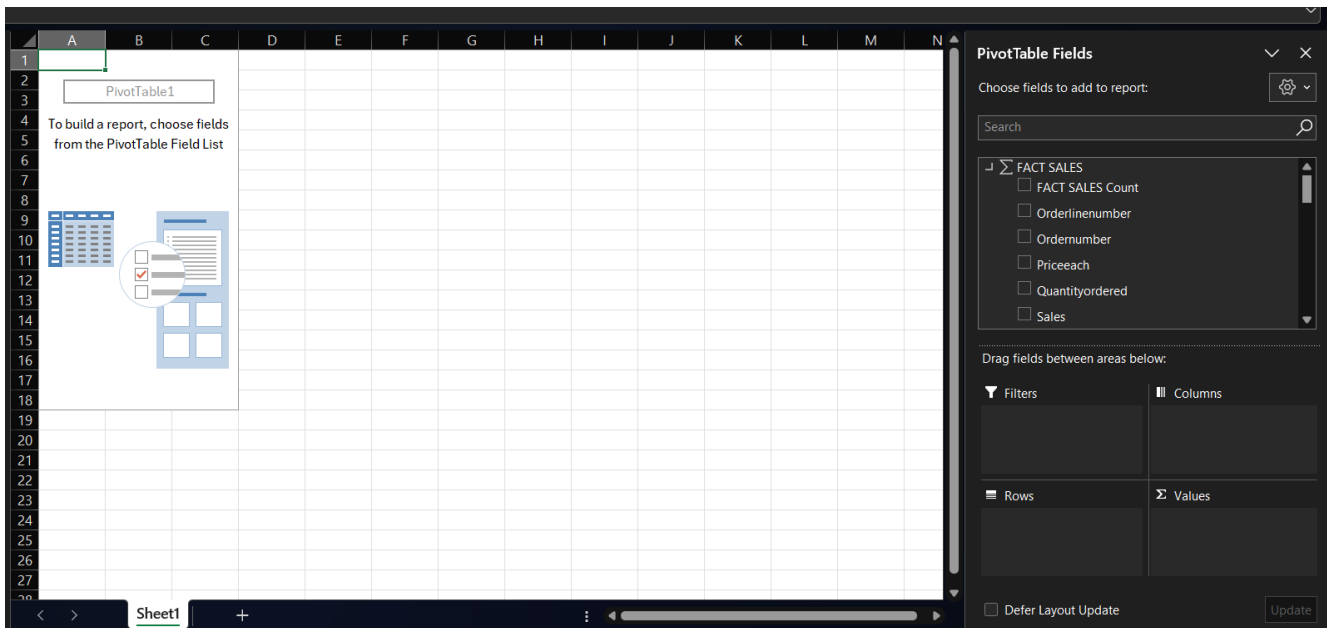
2.3 Deployment

The cube was successfully deployed and processed on the local SSAS server after solving connection and authentication issues.



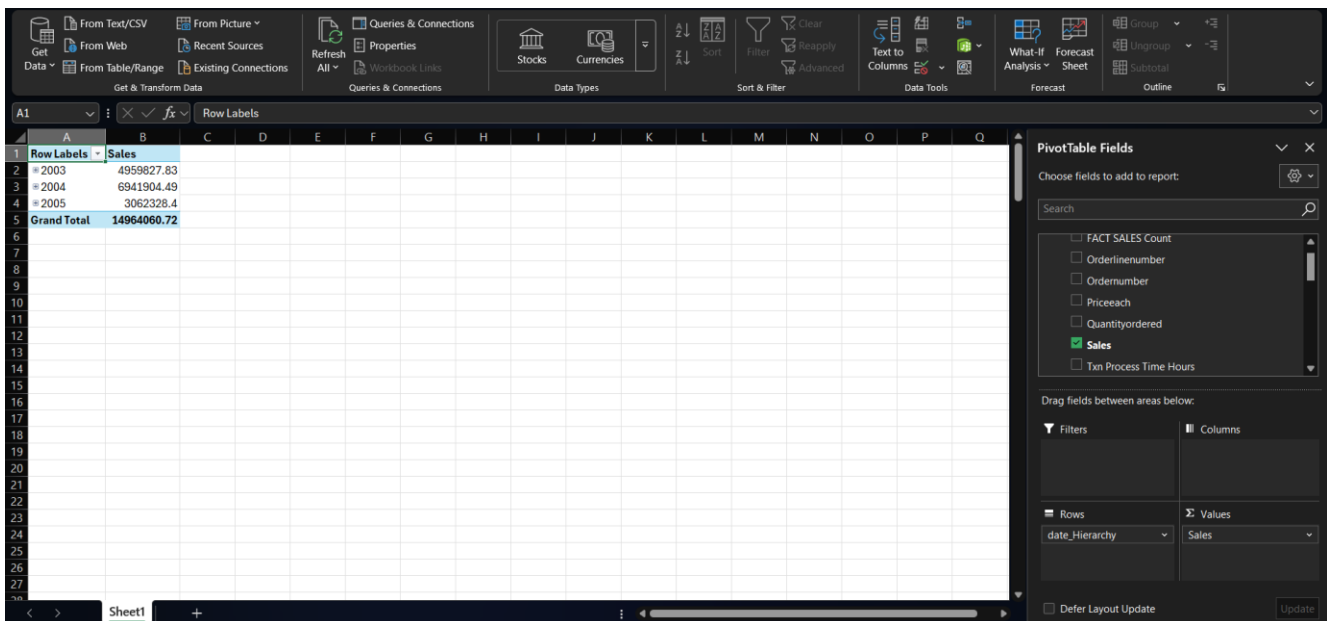
Step 3. Demonstration of OLAP operation

OLAP operations were demonstrated using Microsoft Excel by connecting to the SSAS cube through the analysis services connector. A pivot table was created using the cube data, allowing for Interactive Data analysis. Roll up operations were performed by aggregating data from lower levels such as month to higher levels such as year. Drill down functionality was demonstrated by navigating from year to quarter and month using the defined hierarchy in the date dimension. Slice operations were performed by filtering data based on a single dimension such as selecting a specific year while dice operations involved applying multiple filters such as analyzing sales for a particular product category within a specific region. Pivot operations were carried out by rearranging rows and columns in the pivot table to view data from different perspectives. These operations illustrate the power of OLAP in enabling flexibility and dynamic data analysis.



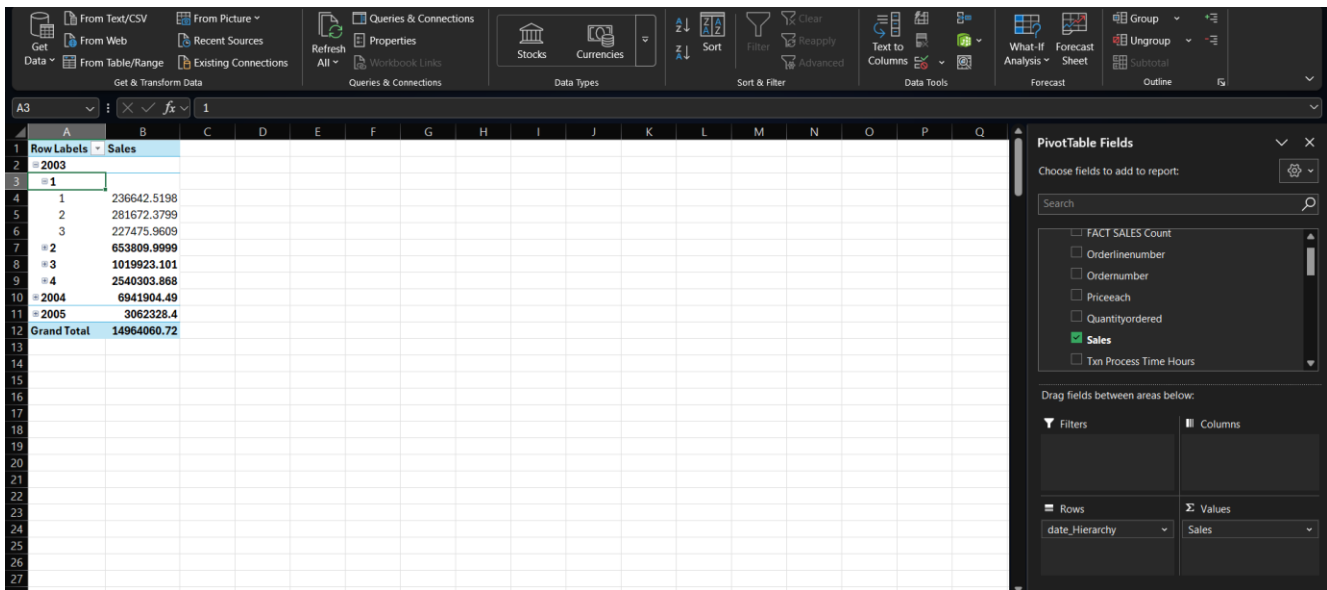
3.1 Roll up

Aggregation of data from month to year value.



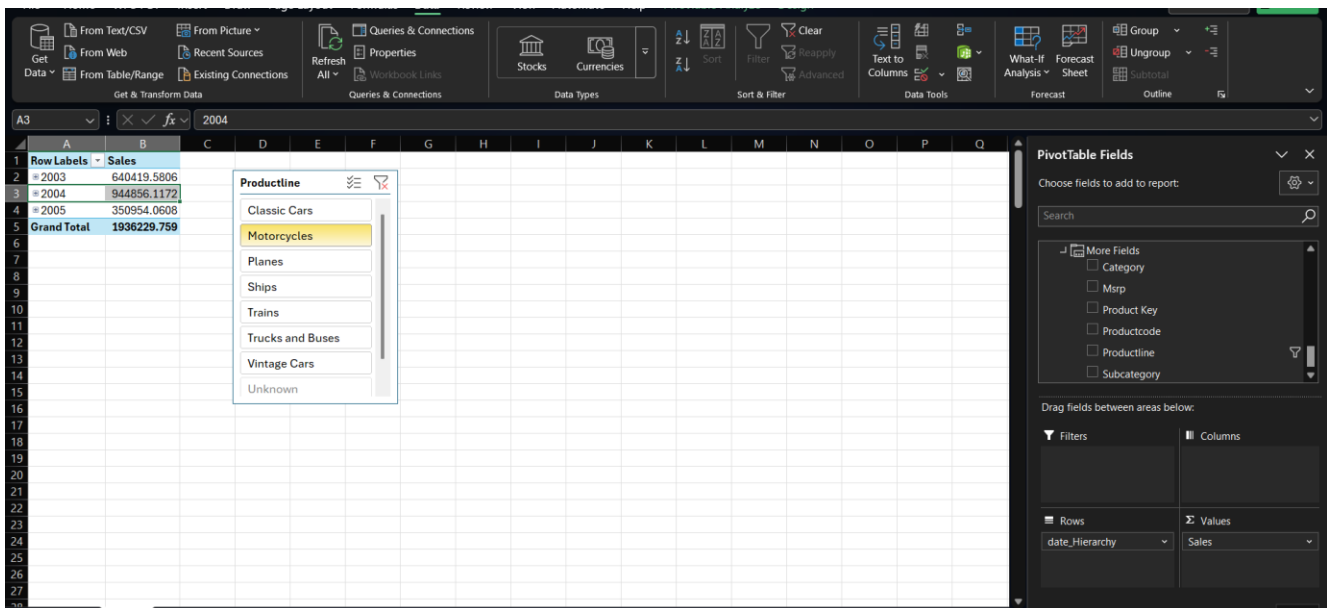
3.2 Drill down

Navigate from year to month.



3.3 Slice

Filters data based on one dimension.



3.4 Dice

Filter data using multiple dimensions. In here country is being used additionally.

The screenshot shows an Excel PivotTable with the following data:

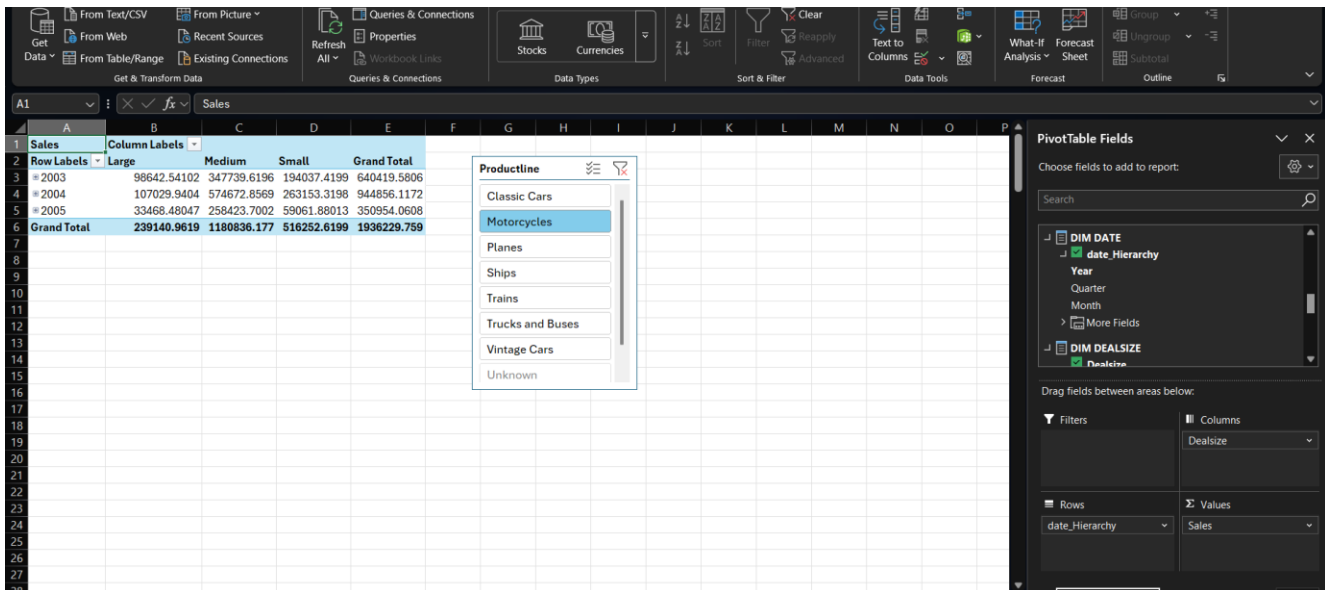
Country	Productline	Sales
Australia	Classic Cars	5561.800049
Australia	Motorcycles	5561.800049
Australia	Planes	5561.800049
Australia	Ships	5561.800049
Australia	Trucks and Buses	5561.800049
Australia	Vintage Cars	5561.800049
Australia	Trains	5561.800049
Australia	Unknown	5561.800049

3.5 Pivot

Change the orientation of data. Rows and columns are swapped.

The screenshot shows an Excel PivotTable with the following data:

Productline	Year	Sales
Classic Cars	2003	98642.54102
Classic Cars	2004	107029.9404
Classic Cars	2005	33468.48047
Classic Cars	Grand Total	239140.9619
Motorcycles	2003	347739.6196
Motorcycles	2004	574672.8569
Motorcycles	2005	258423.7002
Motorcycles	Grand Total	1180836.177
Planes	2003	194037.4199
Planes	2004	263153.3198
Planes	2005	59061.88013
Planes	Grand Total	516252.6199
Ships	2003	640419.5806
Ships	2004	944856.1172
Ships	2005	350954.0608
Ships	Grand Total	1936229.759
Trucks and Buses	2003	
Trucks and Buses	2004	
Trucks and Buses	2005	
Trucks and Buses	Grand Total	
Vintage Cars	2003	
Vintage Cars	2004	
Vintage Cars	2005	
Vintage Cars	Grand Total	
Trains	2003	
Trains	2004	
Trains	2005	
Trains	Grand Total	
Unknown	2003	
Unknown	2004	
Unknown	2005	
Unknown	Grand Total	

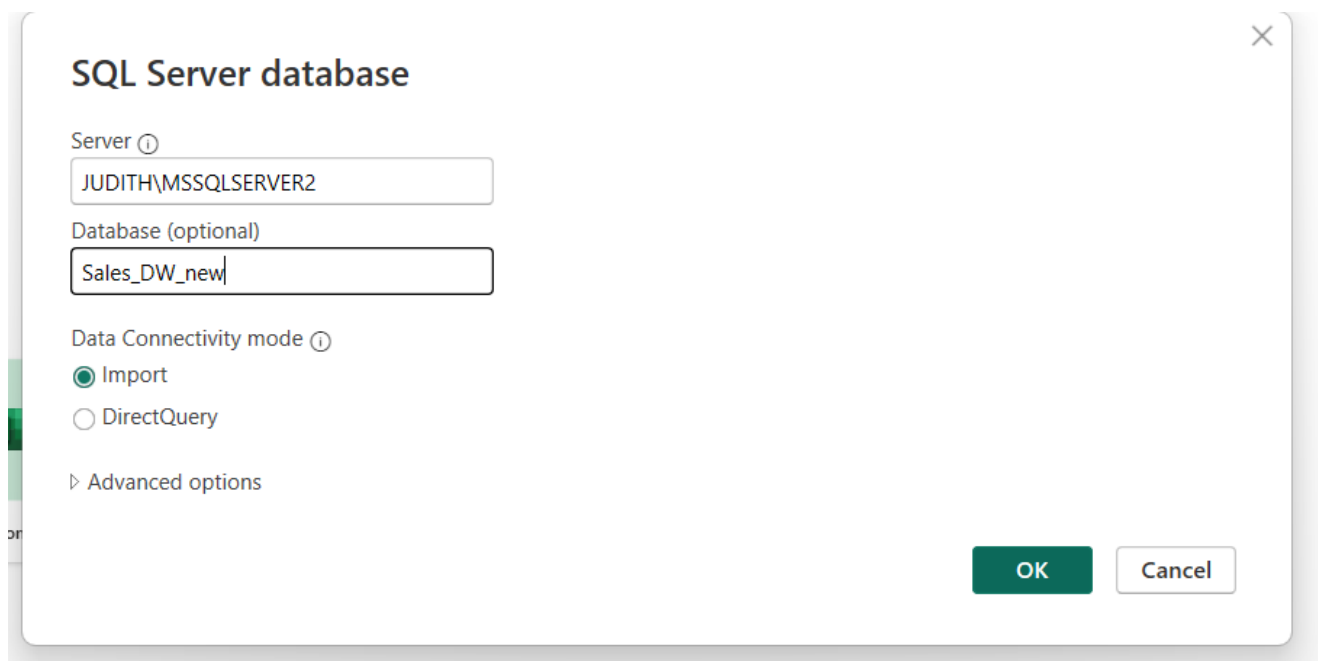
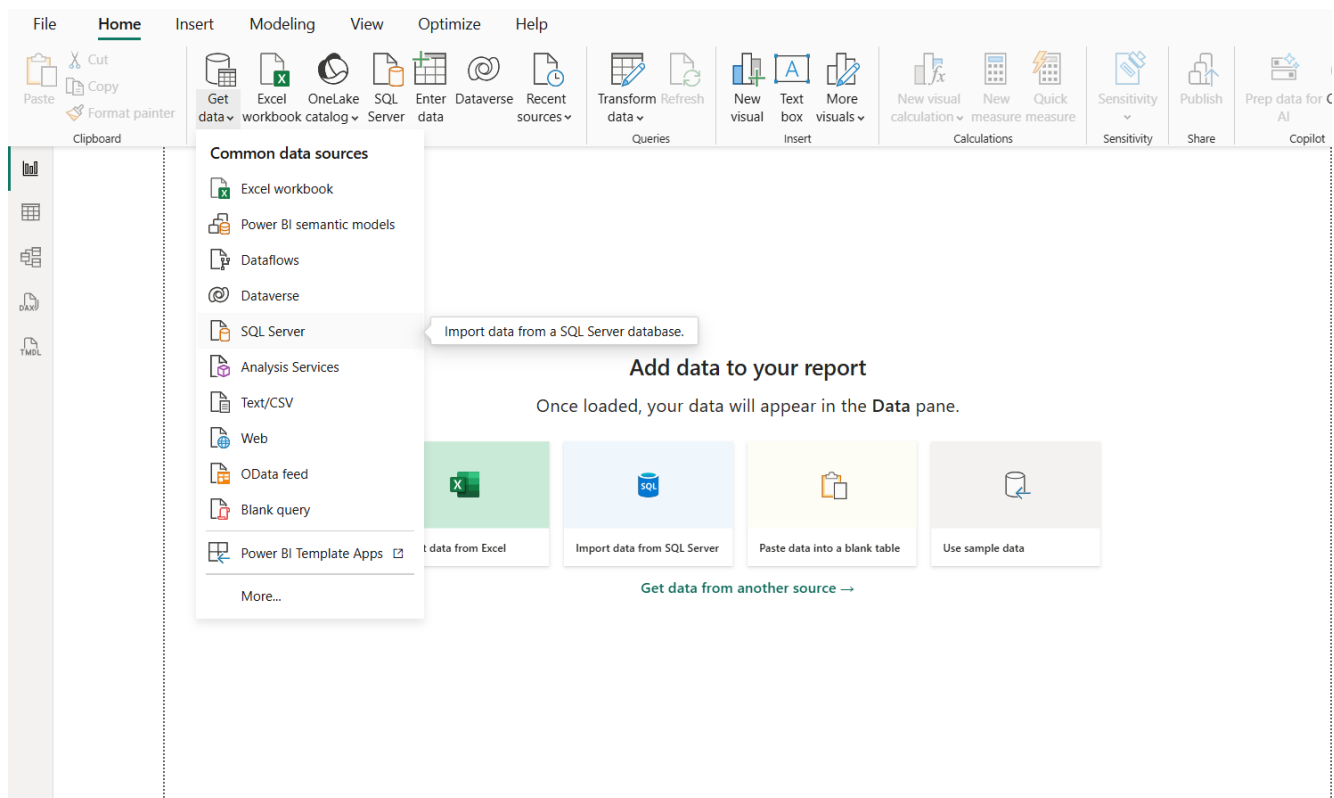


Step 4. PowerBI Reports

Power BI was used to create interactive reports by connecting to the data warehouse through SQL Server using import mode. The required fact and dimension tables were loaded and relationships were established to recreate this task schema within power BI. Multiple reports were developed to demonstrate various analytical capabilities. A matrix visual was created to display detailed tabular data with row and column groupings allowing comparisons across dimensions. A report with multiple slicers were developed to enable filtering, we're selecting a value in one slicer dynamically filters the options in another, demonstrating cascading filters. A drill down report was implemented using a hierarchy of year, quarter and month allowing users to explore data at different levels of detail. Additionally a drill through report was created to enable navigation from a summary visual to a detailed report page based on selected data. The reports were published to power BI service and successfully accessed online.

4.1 Data preparation

Power Bi is connected to the Data warehouse table.



Navigator

Display Options ▾

JUDITH\MSSQLSERVER2: Sales_DW_new [7]

- ☒ DIM_CUSTOMER
- ☒ DIM_DATE
- ☒ DIM_DEALSIZE
- ☒ DIM_PRODUCT
- ☒ **FACT_SALES**
- ☐ sysdiagrams
- ☐ fn_diagramobjects

FACT_SALES

Preview downloaded on Monday

fact_sales_id	ordernumber	orderlinenumber	customer_key	product_key
1	10107	2	1	
2	10121	5	2	
3	10134	2	3	
4	10145	6	4	
5	10168	1	5	
6	10180	9	6	
7	10188	1	7	
8	10201	2	8	
9	10211	14	9	
10	10237	7	11	
11	10251	2	12	
12	10263	2	13	
13	10275	1	14	
14	10285	6	15	
15	10309	5	17	
16	10318	1	18	
17	10329	1	1	
18	10341	9	20	
19	10361	13	21	
20	10375	12	14	
21	10388	4	23	
22	10417	2	25	

Select Related Tables

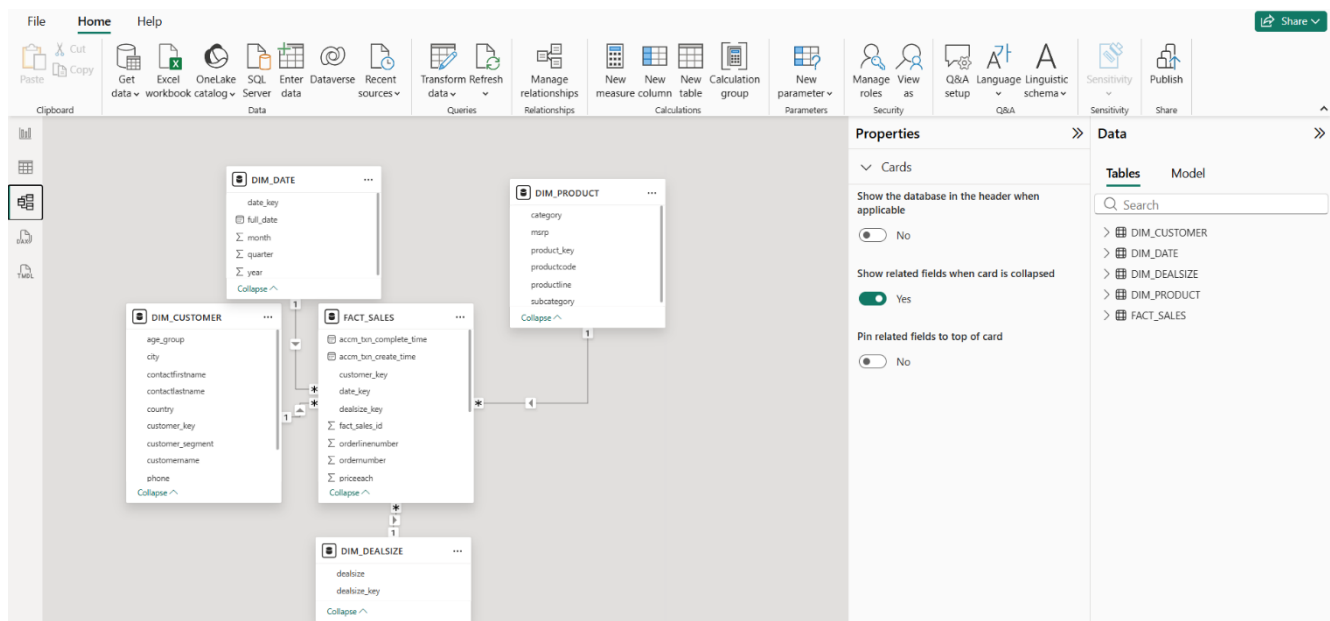
Load

Transform Data

Cancel

4.2 Data modelling

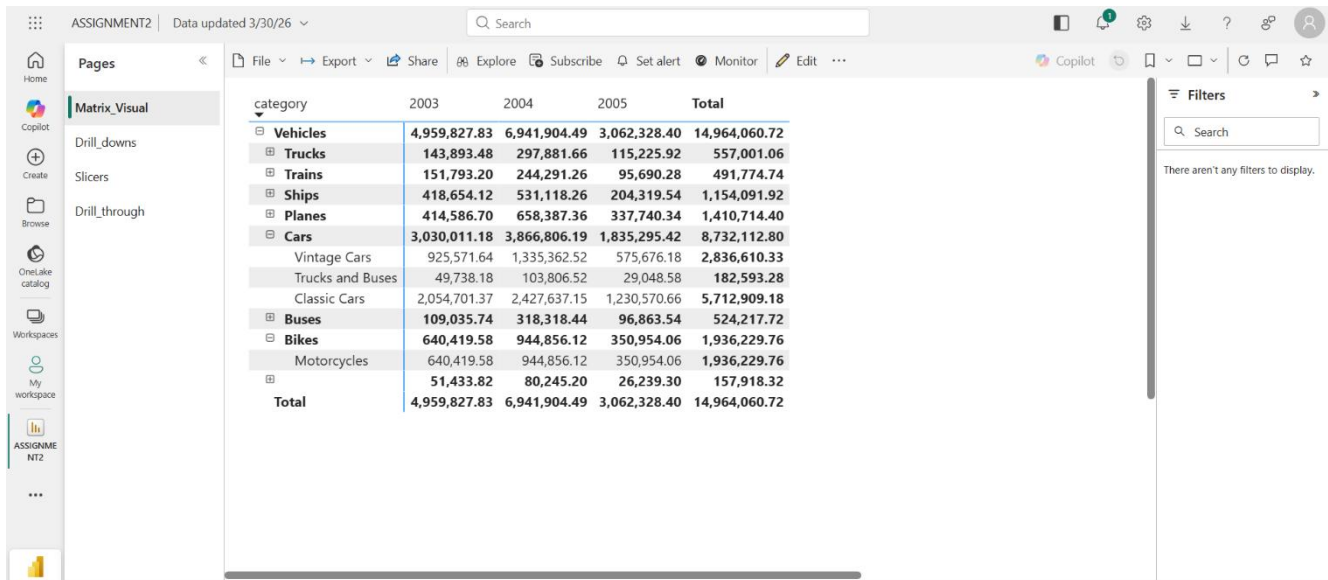
The required fact and dimension tables were loaded and relationships were established to recreate this task schema within power BI.



4.3 Reports

4.3.1 Matrix visual

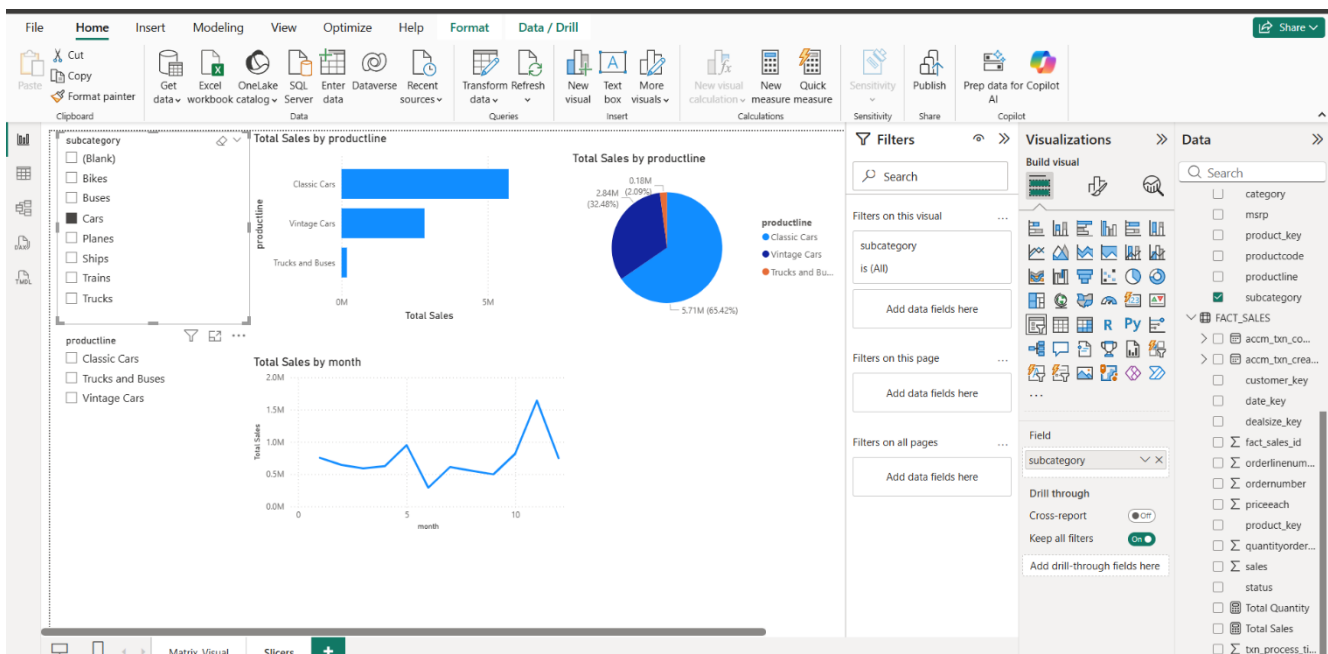
A matrix visual was created to display detailed tabular data with row and column groupings allowing comparisons across dimensions.



category	2003	2004	2005	Total
Vehicles	4,959,827.83	6,941,904.49	3,062,328.40	14,964,060.72
Trucks	143,893.48	297,881.66	115,225.92	557,001.06
Trains	151,793.20	244,291.26	95,690.28	491,774.74
Ships	418,654.12	531,118.26	204,319.54	1,154,091.92
Planes	414,586.70	658,387.36	337,740.34	1,410,714.40
Cars	3,030,011.18	3,866,806.19	1,835,295.42	8,732,112.80
Vintage Cars	925,571.64	1,335,362.52	575,676.18	2,836,610.33
Trucks and Buses	49,738.18	103,806.52	29,048.58	182,593.28
Classic Cars	2,054,701.37	2,427,637.15	1,230,570.66	5,712,909.18
Buses	109,035.74	318,318.44	96,863.54	524,217.72
Bikes	640,419.58	944,856.12	350,954.06	1,936,229.76
Motorcycles	640,419.58	944,856.12	350,954.06	1,936,229.76
Total	4,959,827.83	6,941,904.49	3,062,328.40	14,964,060.72

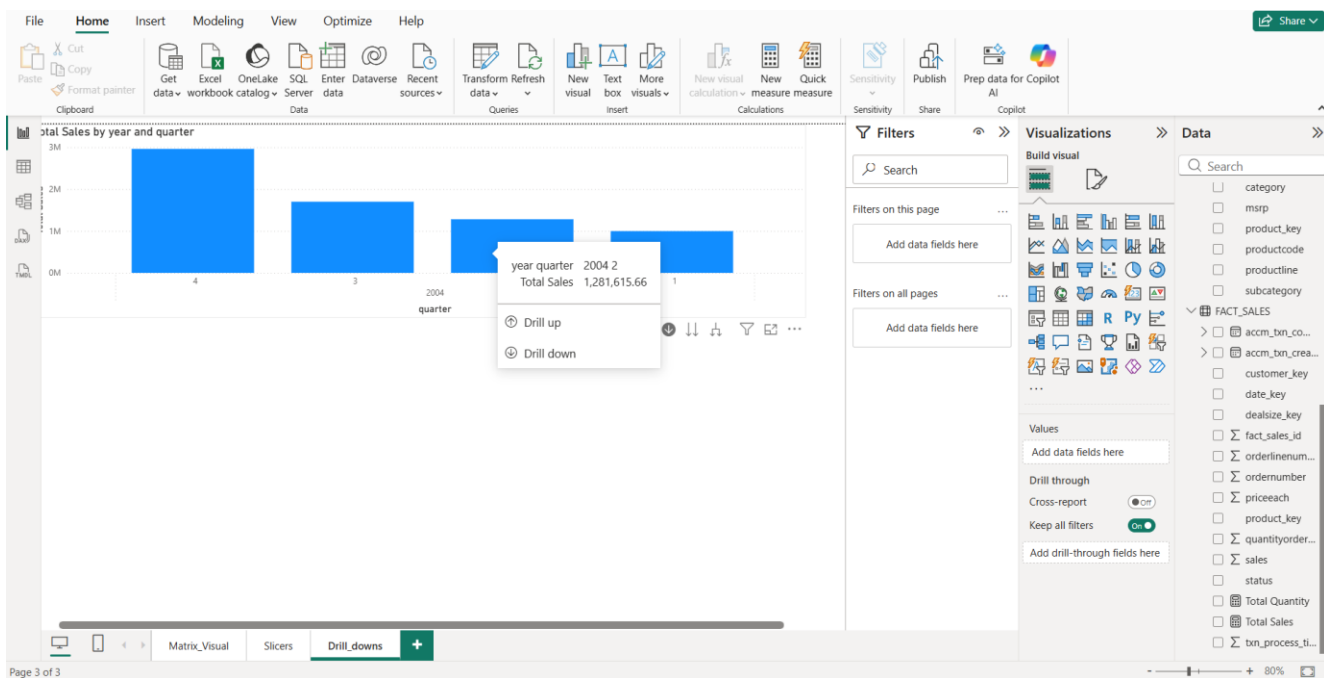
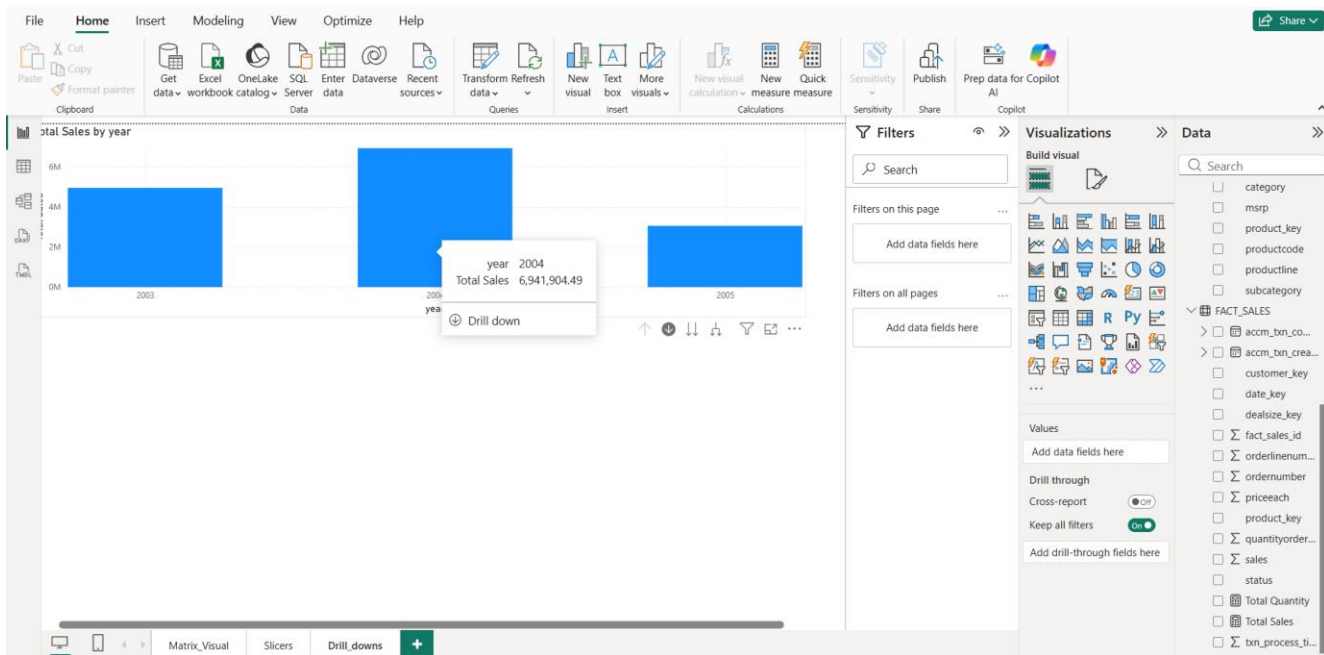
4.3.2 Slicers and cascading filters

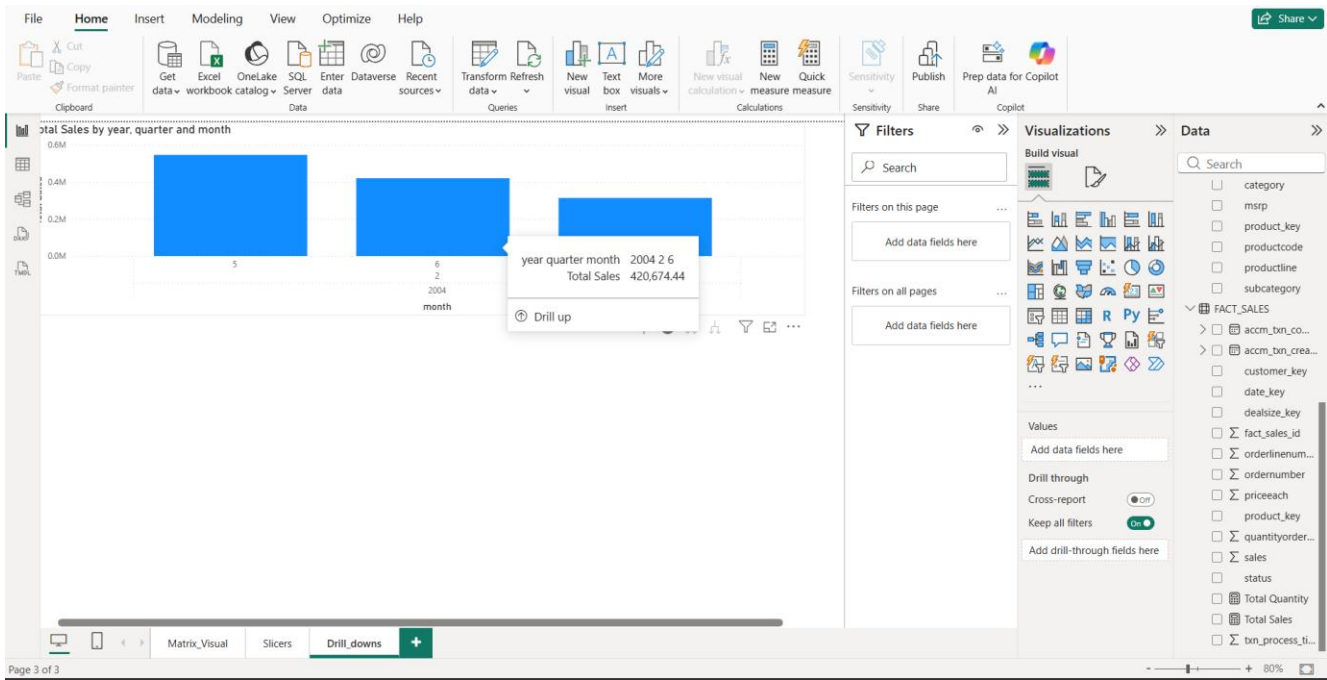
A report with multiple slicers were developed to enable filtering, we're selecting a value in one slicer dynamically filters the options in another, demonstrating cascading filters.



4.3.3 Drill down report

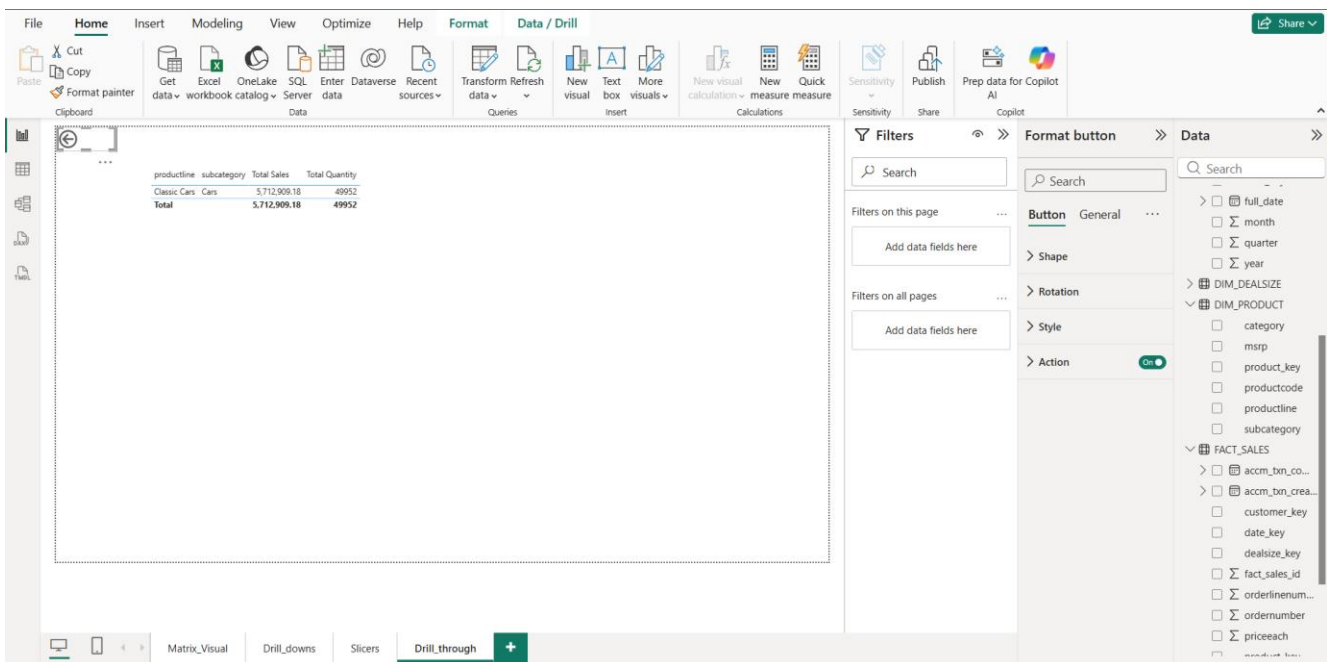
A drill down report was implemented using a hierarchy of year, quarter and month allowing users to explore data at different levels of detail.

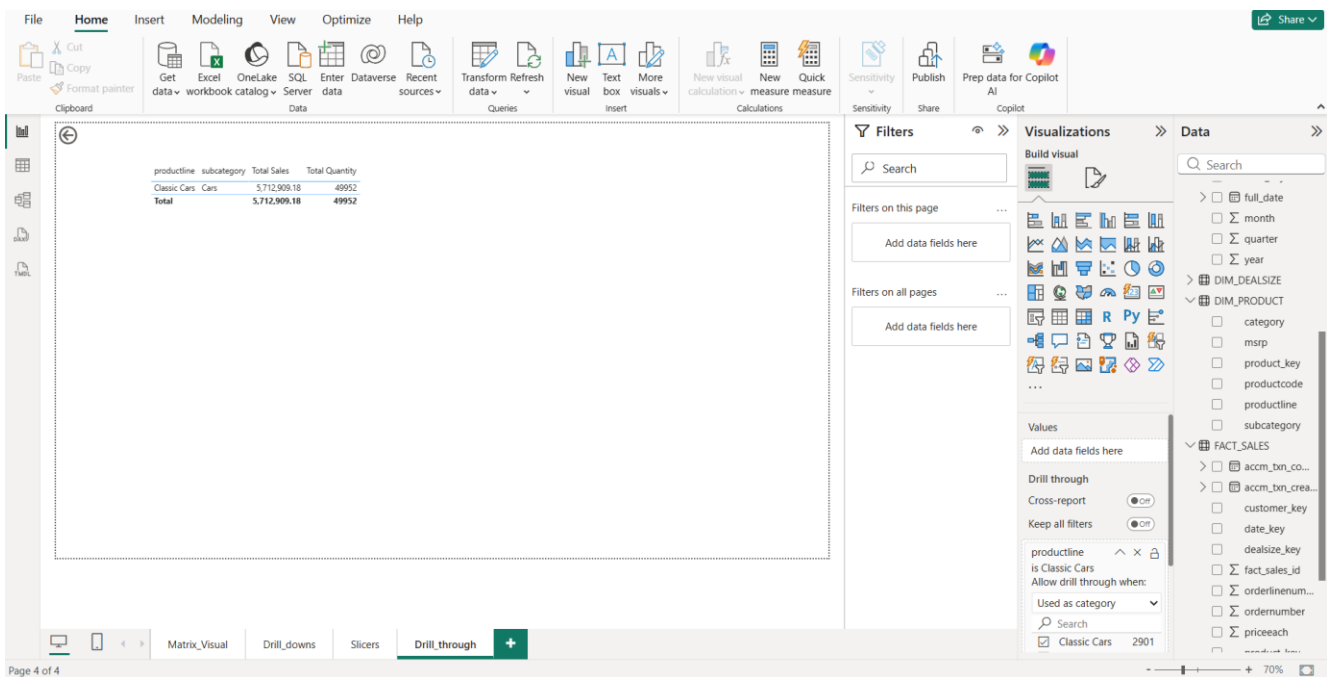
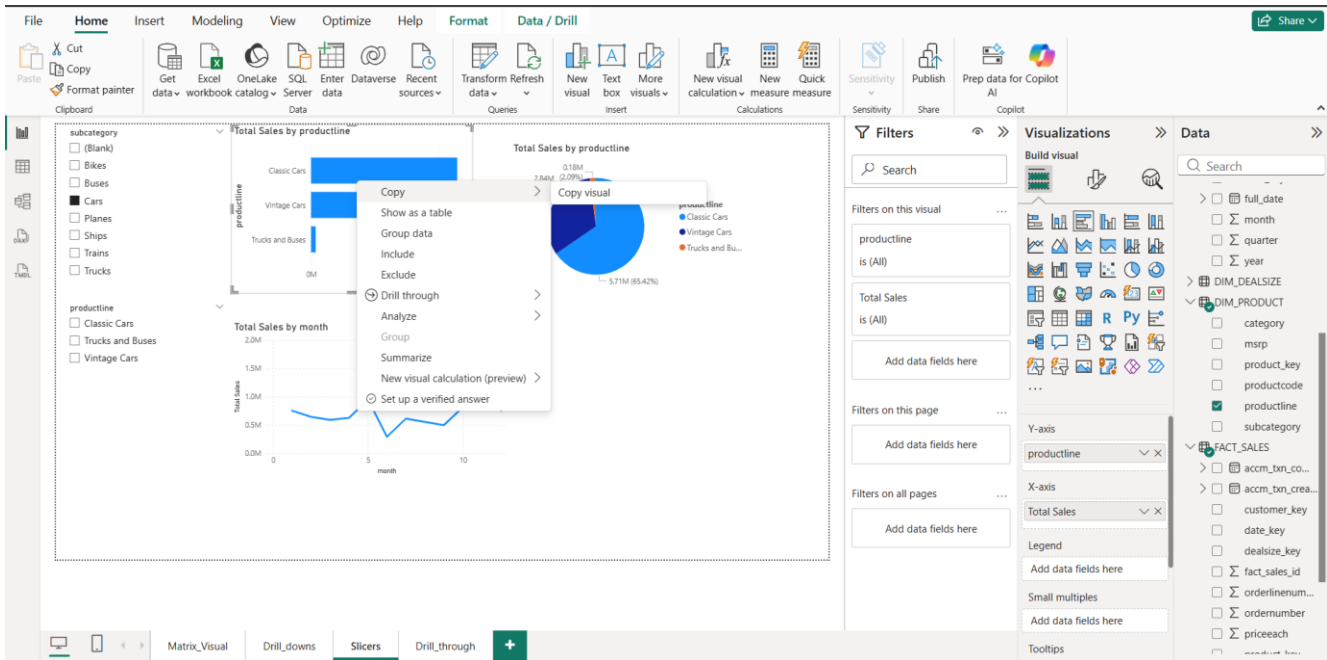




4.3.4 Drill through report

A drill through report was created to enable navigation from a summary visual to a detailed report page based on selected data.





4.4 Publishing

The reports were published to power BI service and successfully accessed online.

Publish to Power BI



Select a destination

Search

My workspace

Select

Cancel

Power BI My workspace

Search



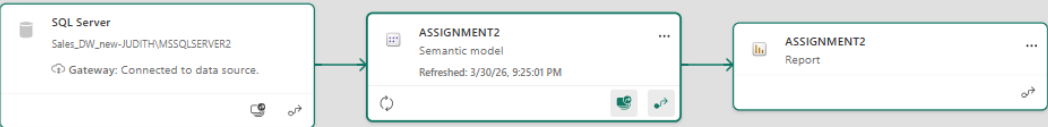
My workspace

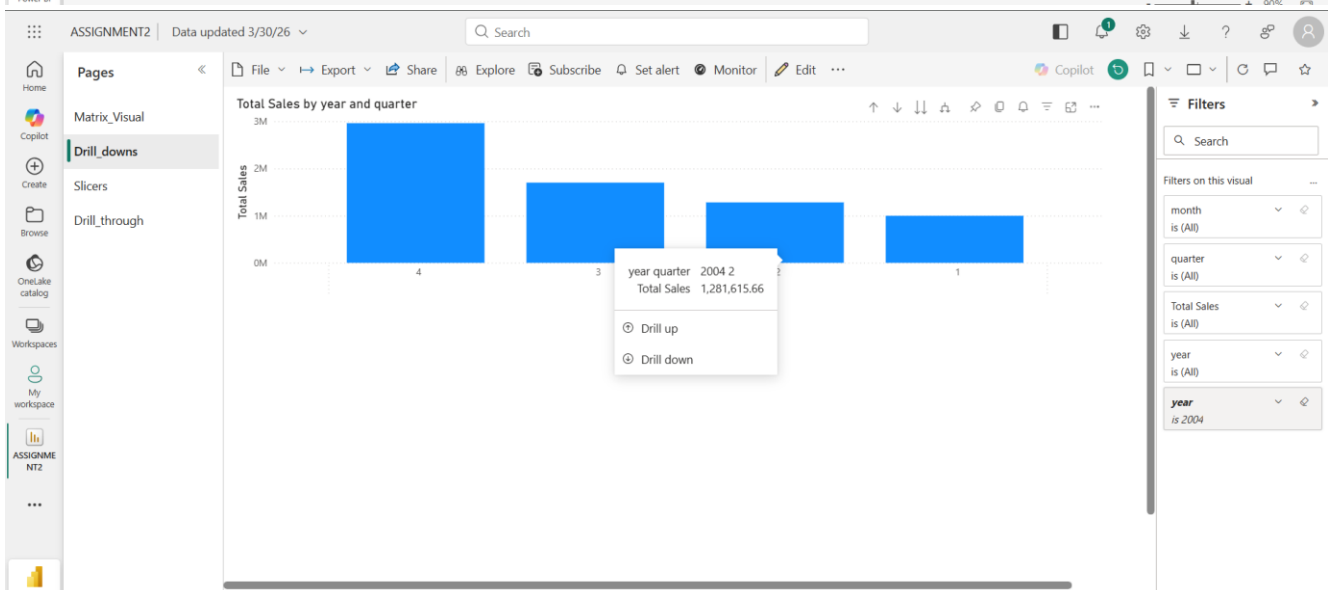
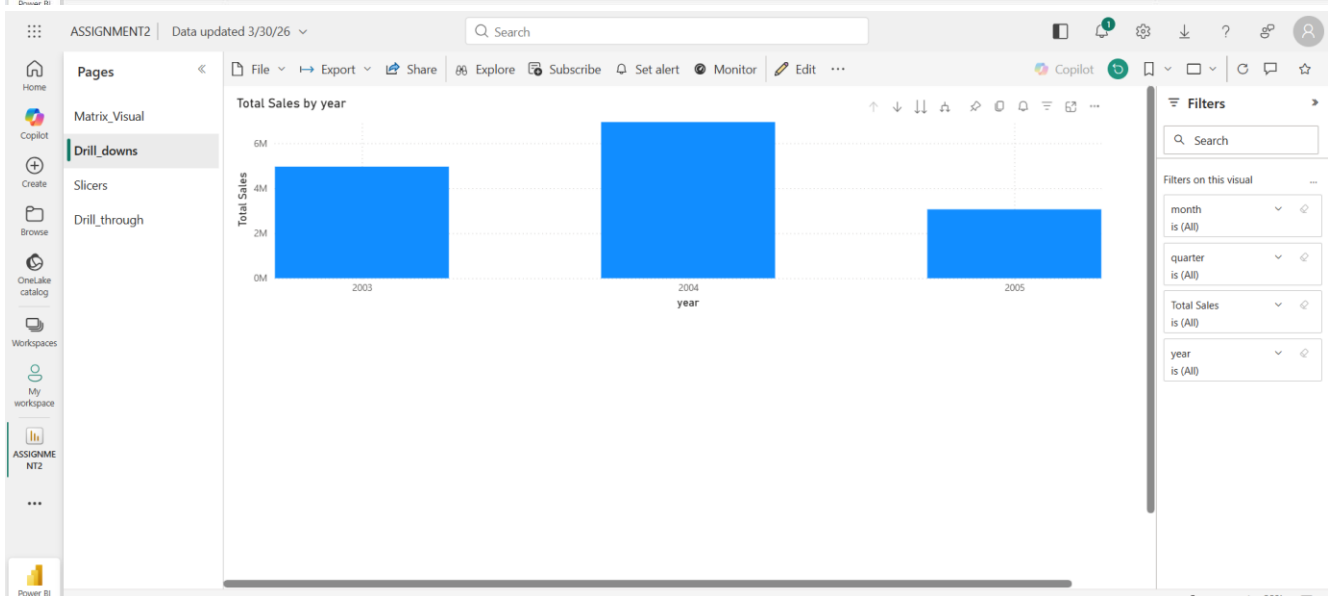
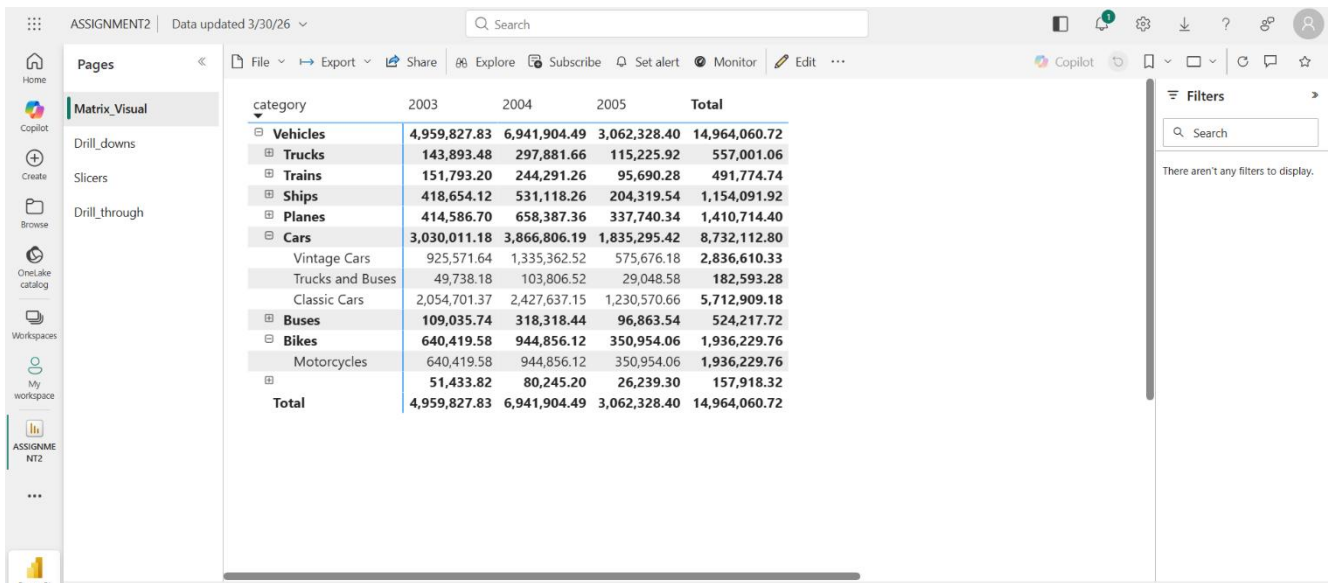
+ New item

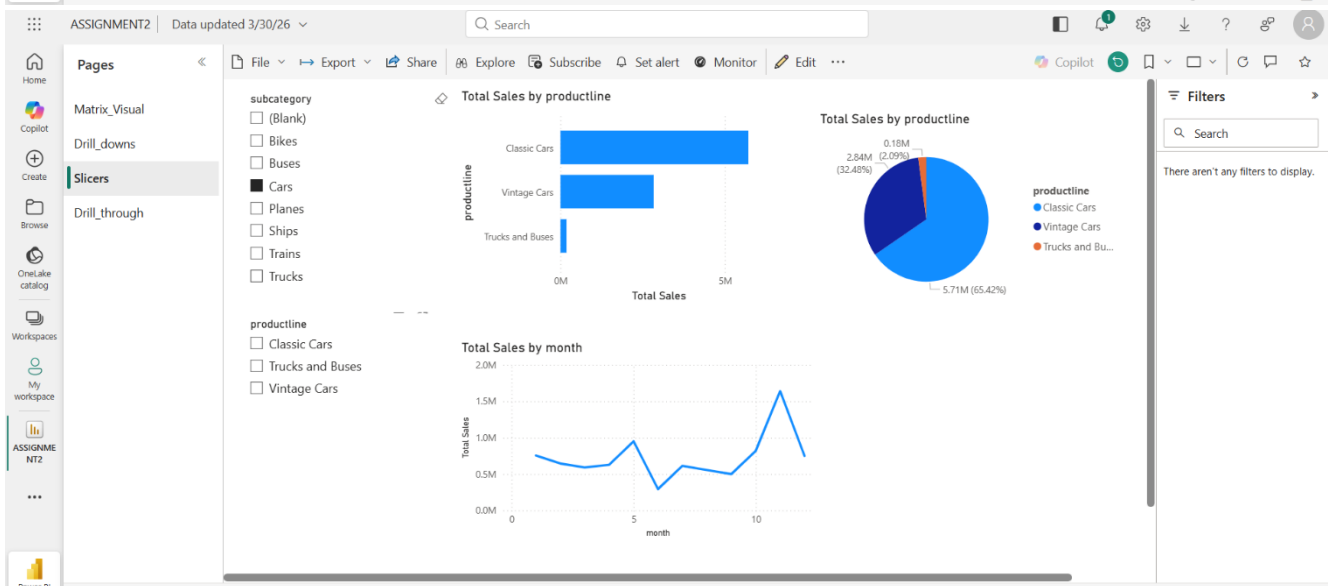
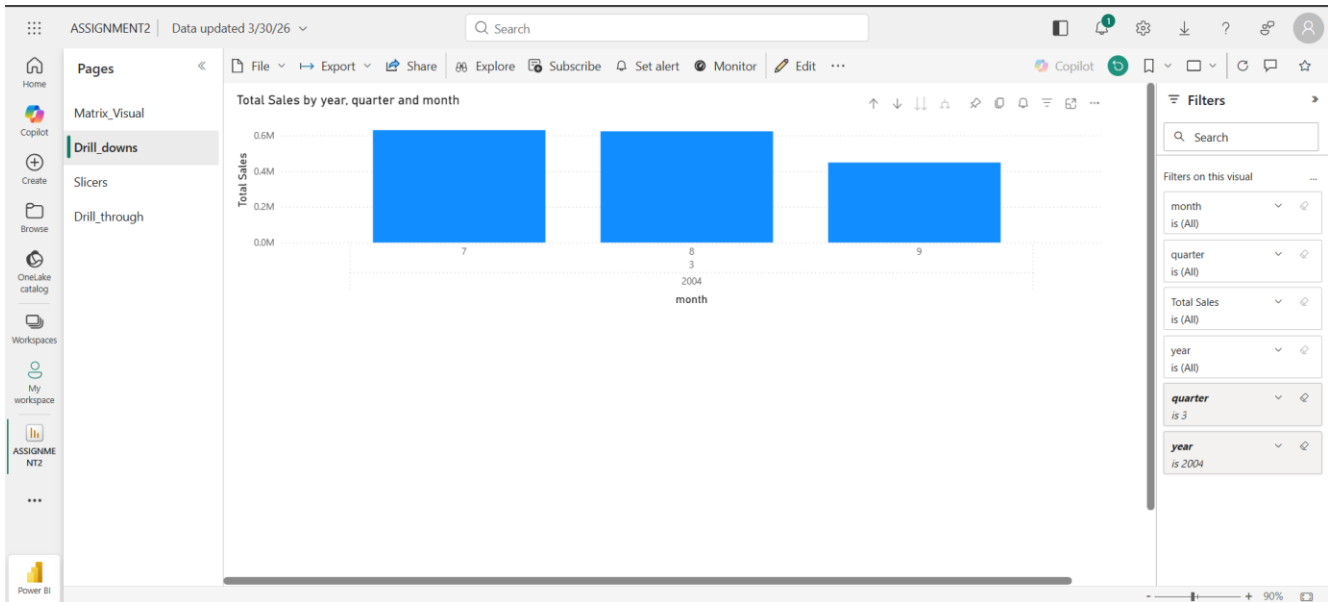
New folder

Import

Migrate







ASSIGNMENT2 | Data updated 3/30/26

Search

Pages: Matrix_Visual, Drill_downs, Slicers, Drill_through

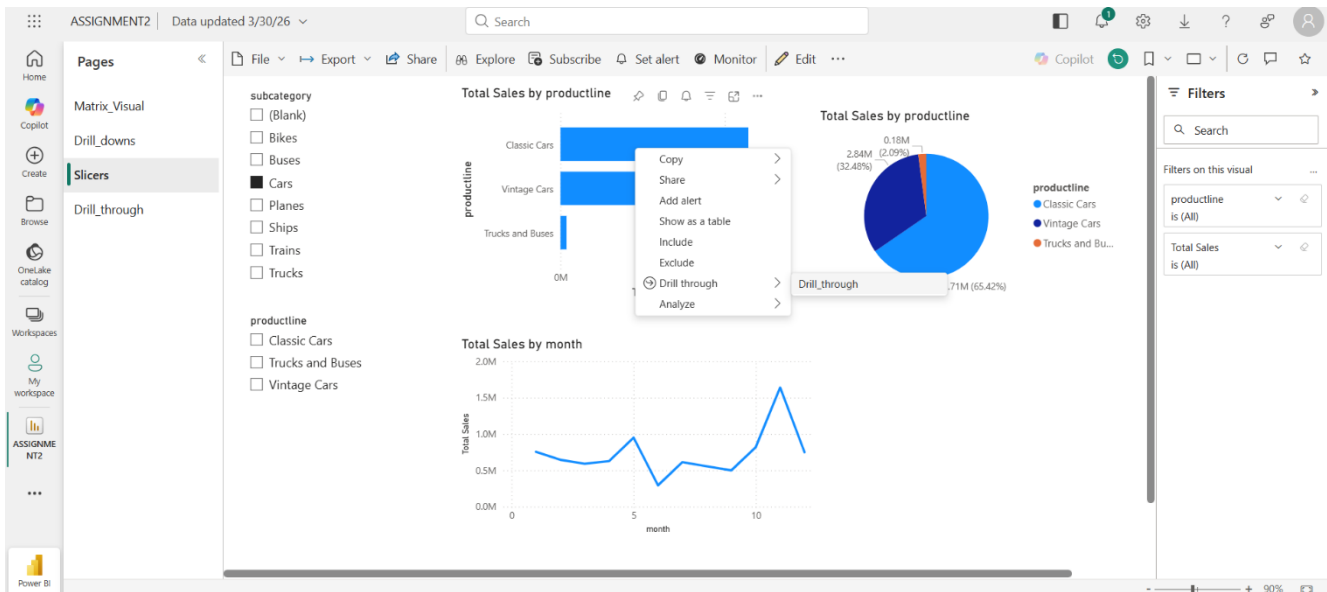
File | Export | Share | Explore | Subscribe | Set alert | Monitor | Edit

Copilot

productline	subcategory	Total Sales	Total Quantity
Classic Cars	Cars	5,712,909.18	49952
Total		5,712,909.18	49952

Filters:

- productline is Classic Cars



Conclusion

This Assignment successfully demonstrated the end to end implementation of a business intelligence solution using a data warehouse, SSAS cube, excel OLAP analysis, and power BI reporting tools. The data warehouse designed in assignment 1 was effectively utilized as the central data source ensuring A structured and consistent foundation for analytical processing.

An SSAS cube was developed by integrating fact and dimension tables, enabling efficient multidimensional analysis. The inclusion of hierarchies enhanced data exploration capabilities and supported meaningful aggregation of measures. Furthermore, OLAP operations such as roll up, drill down, slice, dice and pivot was successfully demonstrated using excel, highlighting the flexibility and power of multidimensional data analysis.

In the visualization phase, power BI was used to create interactive and insightful reports. The matrix report provided structured tabular insights with grouping while the slicer based report enabled dynamic filtering through cascading parameters. The drilldown report allowed hierarchical exploration of time based data, and the drill through report provided detailed context specific analysis through intuitive navigation. These reports collectively improved data accessibility and supported better decision making.

Overall, this assignment provided practical experience in designing and implementing a complete BI workflow, from data modeling to advanced visualization. It strengthened understanding of data warehouse concepts, OLAP techniques, and modern reporting tools, which are essential for real world business intelligence applications.